

Sell my Pet Food

Prabhav Y, Aryaman S, Paarth A



Problem



IGUANA!



HUNGRY IGUANAS :(



Input Files

- Each post starts w/ a specific username
- Algorithm scans each post and checks for iguana-related words like “basking,” “terrarium,” etc

```
classicalGuy "Florida here we come to see the wild iguanas!"
janelz001 "My green iguana loves the humidity in the windy city!"
pet_lover101 "Iguanas are the best! I wish I had a house full of reptiles."
always_an_adventure "I can't believe you didn't help me clean the terrarium."
thestarsareouttonight "Congratulations on your new baby hatchling!"
painted_mountain "Trouble in paradise, my lizard is shedding everywhere."
dannyydandan "Look at how cute he is! #bestiguanamamma!"
mannytheman "I have so many things to say about UVB lighting. Where to start?"
billymait "At any given moment an iguana needs its calcium powder."
willfred "There is no fear of saying that the desert and the jungle cages are di
bobbybob "I did it! I grew the biggest hibiscus flower for my lizard to eat!"
gil_marti "This is music to my ears. Stop scratching at the glass to the beat."
Joeyespin "I cant wait for the expo. Reptiles and amphibians here I come!"
curiouscat "What came first, the iguana or the egg?"
jclubbock "What we see depends mainly on which heat lamp we look for."
wisdon "To understand cold-blooded behavior takes time."
thisiszen "The misting system is the fog one has to see through."
happyperson "There's no greater love than the love of a scaly friend."
peaceforall "Herbivorous lizards make me happy."
dandi_lion "I want a giant lizard for christmas."
martagon "Dandelion greens are beautiful and delicious for my pet."
ocelot "My iguana is so mean, she whipped me with her tail."
puma "I went running today and saw two cute wild iggy lizards."
colocolo "I got my son a baby reptile for christmas. We named him Godzilla."
margay "I can't help but love these little dinosaurs."
serual "Their scales are so colorful."
wwolfie "He's so big he could be a prehistoric monster."
callitwhatitis "She went crazy and ate all the kale in the house!"
annamarie "Clearly this lizard doesn't like hotel rooms."
iamenglish "Check out the basking spot I'm building next week!"
bewise "The hatchlings and the adults are at it again."
highwater "The humidity in this tank seems to be too high."
the_toaster "I'm sorry for burning your mustard greens."
afridege "It's too cold in here for a reptile."
ovenoven "The basking rock is nice and hot in here."
choco "Who is up for some delicious crunchy crickets?"
straydog "I miss my little scaly pup."
disbelief "I can't believe my best friend is a cold-blooded lizard."
moneymoney "If I could buy anything in the world it would be a custom enclosure."
```

socialMediaPosts.txt

```
1 + iguana
2 iguanas
3 reptile
4 reptiles
5 rainforest
6 herbivore
7 herbivorous
8 lizard
9 lizards
10 swimming
11 tail-whipping
12 tail-whip
13 scaly
14 scales
15 greens
16 cold-blooded
17 dewlap
18 tropical
19 arboreal
20 basking
21 head-bobbing
22 head-bob
23 shedding
24 shed
25 terrarium
26 vivarium
27 enclosure
28 uvb
29 heat-lamp
30 basking-rock
31 humidity
32 misting
33 hatchling
34 clutch
35 hibiscus
36 dandelion
37 collards
38 kale
39 mustard-greens
40 calcium-powder
41 iggy
42 zonal
43 ectothermic
44 spikes
45 pouncing
46 climbing
```

targetWords.txt

Output File

```
1 @classicalguy Buy our iguana food! It is the best on the market!
2 @janeIzc001 Buy our iguana food! It is the best on the market!
3 @pet_lover101 Buy our iguana food! It is the best on the market!
4 @always_an_adventure Buy our iguana food! It is the best on the market!
5 @thestarsareouttonight Buy our iguana food! It is the best on the market!
6 @painted_mountain Buy our iguana food! It is the best on the market!
7 @dannyydandan Buy our iguana food! It is the best on the market!
8 @billymait Buy our iguana food! It is the best on the market!
9 @bobbybobob Buy our iguana food! It is the best on the market!
10 @curiouscat Buy our iguana food! It is the best on the market!
11 @wisdon Buy our iguana food! It is the best on the market!
12 @thisiszen Buy our iguana food! It is the best on the market!
13 @happyyperson Buy our iguana food! It is the best on the market!
14 @peaceforall Buy our iguana food! It is the best on the market!
15 @dandi_lion Buy our iguana food! It is the best on the market!
16 @martagon Buy our iguana food! It is the best on the market!
17 @ocelot Buy our iguana food! It is the best on the market!
18 @puma Buy our iguana food! It is the best on the market!
19 @colocolo Buy our iguana food! It is the best on the market!
20 @serual Buy our iguana food! It is the best on the market!
21 @callitwhatitis Buy our iguana food! It is the best on the market!
22 @annamarie Buy our iguana food! It is the best on the market!
23 @iamenglish Buy our iguana food! It is the best on the market!
24 @bewise Buy our iguana food! It is the best on the market!
25 @highwater Buy our iguana food! It is the best on the market!
26 @the_toaster Buy our iguana food! It is the best on the market!
27 @afridge Buy our iguana food! It is the best on the market!
28 @ovenoven Buy our iguana food! It is the best on the market!
29 @straydog Buy our iguana food! It is the best on the market!
30 @disbelief Buy our iguana food! It is the best on the market!
31 @moneymoney Buy our iguana food! It is the best on the market!
32
```

iguanaAd.txt

- The users who mentioned iguana-related terms were the only people who received this advertisement
- Distinction example:
 - “Dogs are the best pet”
 - Ad **ISN’T** targeted towards them
 - “My iguana loves swimming”
 - Ad **IS** targeted towards them
 - People w/o a pet reference aren’t targeted either

Algorithm 1 (Data Parsing)

```
// loop through the posts and target words to find the users that will receive the ad
String post = data.getNextPost();
while (!post.equals("NONE")) {
    // word before space is username -> find username by finding space and then using substring to get the username
    int findSpace = post.indexOf(" ");
    String findUser = post.substring(0, findSpace);
    // check if post contains any of the target words, if it does then add the user to the users string
    boolean target = false;
    String targetWord = data.getNextTargetWord();
    while (!targetWord.equals("NONE")) {
        // paarth add .lower please, check if targetWord exists in post
        if (post.indexOf(targetWord) != -1) {
            target = true;
        }
        targetWord = data.getNextTargetWord();
    }
    // if theres a matched word then the username will get added to the users string
    if (target) {
        users += findUser + " ";
    }
    // move to next post & repeat till end
    post = data.getNextPost();
}
```

- Username is extracted one by one from each post
- Compare post against every target word
- **KEY CODE PARTS:**
 - “indexOf(targetWord) != -1”
 - Checks if word is in post
 - Nested while loop
 - Check every target word for each post
 - “boolean target = false”
 - Doesn’t allow users to be added to multiple posts
- Filter system tailored to iguana related terms

Algorithm 2 (File Creation)

```
// prepping advertisement file where users will be displayed along w/  
data.prepareAdvertisement("iguanaAd.txt", users, "Buy our iguana food! It is the best on the market!");
```

```
// if theres a matched word then the username will get added to the users string  
if (target) {  
    users += findUser + " ";  
}
```

- Built one large string of usernames separated by spaces
- DataCollector splits the string and writes each user on a new line w/ the ad

```
for (String un : usernames.split(" "))  
{  
    | fw.write("@ " + un + " " + advertisement + "\n");  
}
```

- “split(“ ”) separates the usernames
- Each username gets:
 - “@username + ad”
- This happens in the “iguanaAd.txt” file

Development Process and Milestones

Feature	Estimated Time	Actual Time
File Reading	15 min	20 min
Loop through posts	30 min	1 hr
Matching target words	30 min	55 min
Output file	20 min	1 hr

Biggest Challenge:

- Not knowing why target words weren't properly resetting
- getNextTargetWord() resets index to zero after finishing
- Words would have been skipped if the method didn't work
- Overcame by: implementing logic within getNextTargetWord() to manually reset the index to zero upon completion

Ethical Reflection

- This same algorithm is used by major tech companies
 - Instagram
 - TikTok
 - YouTube
- Other large companies as well
 - News organizations: push their own agendas
 - Sports gambling: DraftKings, FanDuel: keep users addicted
 - Consumer brands: pushing their products out
- All of these companies track specific keyword and target ads
- **Discussion Question:** If a user only receives targeted news and never searches for anything themselves, what are the ethical/social implications of having every article chosen for them by an algorithm?
 - Filter bubbles: users only see information which confirms their existing beliefs
 - Echo chambers: critical thinking prevented when no opposing views are introduced
 - Mass manipulation: whoever controls the algorithm can slowly shift public opinion
- The result?
 - Increased political polarization
 - Increased consumerism
 - Lack of genuine connection between different people

Conclusion

- Our final solution efficiently processes many posts using nested loops and string parsing
- Makes targeting ads automatic and increase market precision

